

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling

standards

(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I D.Uday Kiran(192425161) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D.Uday Kiran

Comparative Analysis

1. Star Topology vs Bus Topology

Question 1: Network Connectivity for a Remote Rural Branch Office

Understanding of Problem & Constraints

The multinational company is establishing a branch office in a remote rural location where high-speed wired Internet is unavailable. Therefore, the network solution must overcome the limitations of the location while supporting important business activities.

The major constraints are:

High-speed wired Internet is unavailable.

The office needs sufficient bandwidth for cloud applications.

Video conferencing requires stable and low-latency connectivity.

Secure communication with headquarters is essential.

The connectivity budget is limited.

Business operations should continue even if the primary Internet connection fails.

The network should be scalable for future employees and applications.

Security must be maintained because company information may be transmitted over the Internet.

The solution should therefore balance bandwidth, reliability, cost, security, and scalability.

Problem Decomposition

The problem can be divided into the following sub-problems:

Select a suitable Internet connection for the rural location.

Provide backup connectivity to reduce downtime.

Select appropriate networking devices.

Design secure communication with headquarters.

Protect the office network from external attacks.

Provide sufficient bandwidth for cloud applications and video conferencing.

Ensure that the solution can support future expansion.

Control installation and operational costs.

This decomposition helps the network engineer select a solution that satisfies all constraints rather than focusing only on Internet speed.

Application of Constraints in Solution

Internet Connection

The primary recommendation is 4G/5G fixed wireless access, if reliable cellular coverage is available at the location. A 5G connection can provide higher bandwidth and lower latency where coverage exists. If 5G is unavailable, 4G LTE can be used.

If cellular connectivity is unreliable or unavailable, satellite Internet can be used as the primary connection or as a backup. Modern low-Earth-orbit satellite services can provide substantially better latency than traditional geostationary satellite connections.

A practical solution is therefore:

Primary connection → 5G/4G fixed wireless

Backup connection → Satellite or second cellular provider

This provides connectivity without requiring expensive wired infrastructure.

Networking Devices

The office should use:

4G/5G modem or outdoor fixed-wireless receiver

Dual-WAN router

Managed Ethernet switch

Wireless access points

Firewall/security gateway

UPS for critical networking equipment

VPN-capable router or firewall

A dual-WAN router is especially useful because it can automatically switch to the backup Internet connection if the primary connection fails.

Security Mechanisms

Secure communication with headquarters should be implemented using a site-to-site VPN, preferably using modern encrypted VPN technologies such as IPsec or WireGuard where supported by the organization's infrastructure.

Other security mechanisms should include:

Firewall

VPN encryption

Strong administrator authentication

Multi-factor authentication

WPA3 for Wi-Fi

Separate guest Wi-Fi

Network segmentation

Regular firmware updates

Intrusion detection/prevention where practical

Centralized logging

The office network should never directly expose internal systems to the public Internet.

Logical Reasoning & Strategy

The most important constraint is the lack of wired broadband. Installing a dedicated wired connection in a remote location may be expensive and time-consuming. Therefore, wireless connectivity is more practical.

Using 5G/4G as the primary connection keeps infrastructure costs relatively low. However, relying on only one wireless provider creates a single point of failure. Therefore, a second connection should be provided.

The dual-WAN router can monitor the primary connection. If the primary connection becomes unavailable, the router can automatically switch traffic to the backup connection.

For example:

Normal operation:

Office → 5G Router → Firewall → Internet → Cloud/Headquarters

Failure condition:

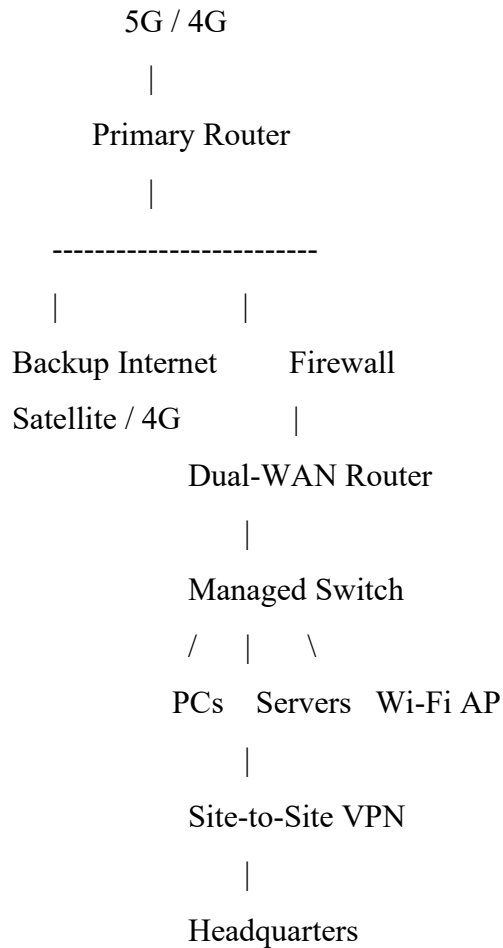
Office → Backup Satellite/4G → Firewall → Internet → Cloud/Headquarters

This strategy improves business continuity without requiring two expensive dedicated wired connections.

Bandwidth management should also be implemented. Video conferencing and critical cloud applications can be prioritized using Quality of Service (QoS). Non-critical activities such as large downloads or recreational streaming can receive lower priority.

Solution Accuracy & Feasibility

The recommended architecture is:



The firewall should enforce security policies, while the VPN provides encrypted communication with headquarters.

The UPS should keep the router, firewall, switch, and access points operating during short power interruptions. This is particularly important in remote locations where power reliability may be lower.

Creativity & Innovation

A cost-effective approach is to use SD-WAN or intelligent multi-WAN routing if the company can justify the additional management capability. This can dynamically select the best available link based on latency, packet loss, and availability.

Another useful approach is local caching and cloud optimization for frequently accessed resources. This can reduce unnecessary bandwidth consumption.

The company can also start with 4G/5G and upgrade to another broadband technology later if wired infrastructure becomes available.

Clarity of Presentation

Requirement Recommended Solution

Primary Internet 5G/4G Fixed Wireless

Backup Internet Satellite or second cellular provider

Routing Dual-WAN Router

Security Firewall + VPN

Wi-Fi WPA3-enabled Access Points

Headquarters Connection Site-to-Site VPN

Reliability Dual WAN + UPS

Performance QoS and bandwidth management

Future Expansion Managed switch and scalable wireless

Cost Control Wireless-first infrastructure

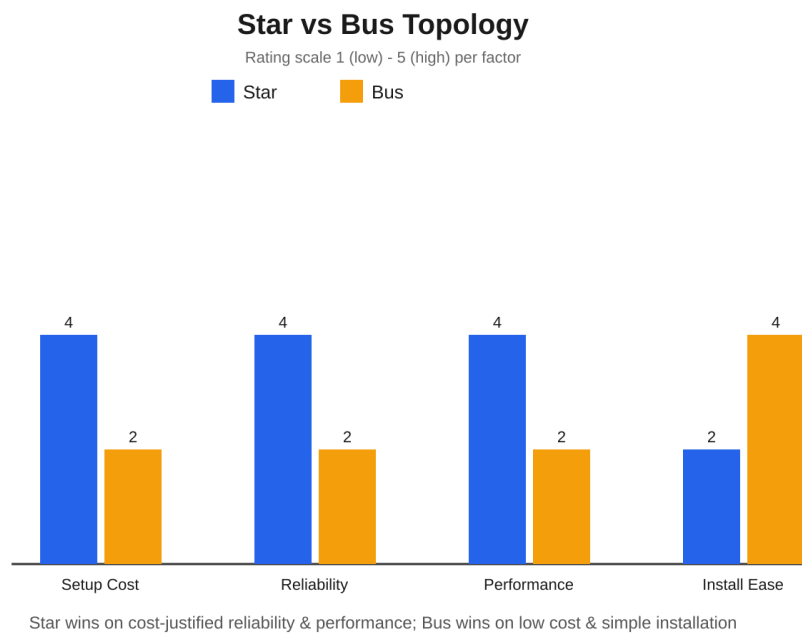


Figure 1: Relative rating of Star vs Bus across cost, reliability, performance, and installation ease

2. Mesh Topology vs Star Topology — Fault Tolerance

Question 2: Wireless Network Design for a University with More Than 3,000 Students

Understanding of Problem & Constraints

The university needs campus-wide wireless access for more than 3,000 students across classrooms, laboratories, libraries, and hostels.

The major constraints are:

More than 3,000 potential users.

Many simultaneous wireless connections.

Large physical campus area.

Different buildings and room layouts.

High-density areas such as classrooms and libraries.

Limited number of access points due to budget constraints.

Signal interference must be minimized.

Users need secure authentication.

The network must provide sufficient performance.

Coverage must be maintained without excessive equipment.

The challenge is therefore not simply to provide Wi-Fi coverage. The design must provide high user capacity, efficient frequency usage, security, and reasonable cost.

Problem Decomposition

The problem can be divided into:

Estimate user density in different locations.

Identify high-density and low-density areas.

Determine appropriate access point locations.

Select suitable frequency bands.

Reduce channel interference.

Select authentication mechanisms.

Manage thousands of users.

Provide roaming between access points.

Prioritize important applications.

Monitor and optimize the wireless network.

Application of Constraints in Solution

Access Point Placement

Access points should not simply be placed at equal distances across the campus. Placement should be based on user density, building structure, walls, interference, and expected traffic.

High-density areas such as:

Lecture halls

Libraries

Laboratories

Student common areas

should receive priority.

Hostels should have access points distributed according to floor layout and expected student density.

The university should conduct a wireless site survey before finalizing AP positions. This helps identify dead zones and interference.

Instead of using maximum transmission power everywhere, AP power should be adjusted so that cells overlap appropriately for roaming without creating excessive interference.

Frequency Band Selection

The university should use a combination of 2.4 GHz, 5 GHz, and, where supported, 6 GHz Wi-Fi.

2.4 GHz

2.4 GHz provides better range and penetration through obstacles but has fewer non-overlapping channels and is more susceptible to interference.

Therefore, it should mainly be used for devices that require longer range or compatibility with older equipment.

5 GHz

5 GHz provides more usable channels and generally supports higher capacity than 2.4 GHz. It should be the primary band for student laptops, smartphones, and high-bandwidth applications where device support is available.

6 GHz

If the university deploys compatible Wi-Fi 6E/7 equipment and client devices, 6 GHz can provide additional spectrum and reduce congestion from older Wi-Fi devices. Its shorter range means careful AP placement is required.

Authentication Methods

A university network should avoid using a single shared Wi-Fi password for thousands of students.

A better solution is WPA2-Enterprise or WPA3-Enterprise with 802.1X authentication.

A centralized authentication server such as RADIUS can verify student credentials.

The authentication process can be:

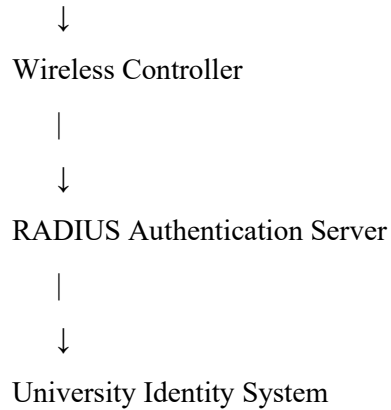
Student Device

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Wireless Access Point

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This allows individual users to authenticate using their university credentials.

When a student leaves the university, the account can be disabled without changing the password for everyone.

Logical Reasoning & Strategy

Because the university has a limited AP budget, simply increasing the number of access points is not possible.

Therefore, the solution should maximize the capacity of each access point.

High-density locations should receive APs first. The university should avoid placing APs solely based on physical coverage because one AP covering a large area may become overloaded when hundreds of students connect simultaneously.

The design should use:

Band steering

Load balancing

Appropriate channel planning

Transmit-power control

Airtime fairness

Fast roaming where supported

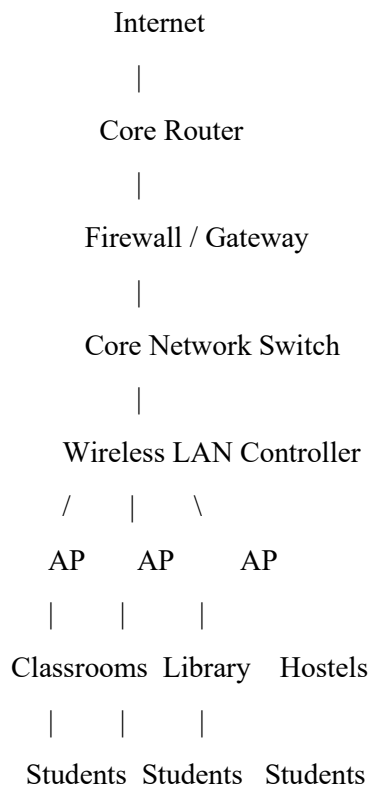
QoS

Centralized wireless management

The 2.4 GHz band should be used carefully because of its limited channel availability. More users should be encouraged to use 5 GHz or 6 GHz where possible.

Solution Accuracy & Feasibility

A suitable architecture is:



The university can divide wireless access into different logical networks:

Student Wi-Fi

Faculty Wi-Fi

Administrative Wi-Fi

Guest Wi-Fi

IoT/Laboratory devices

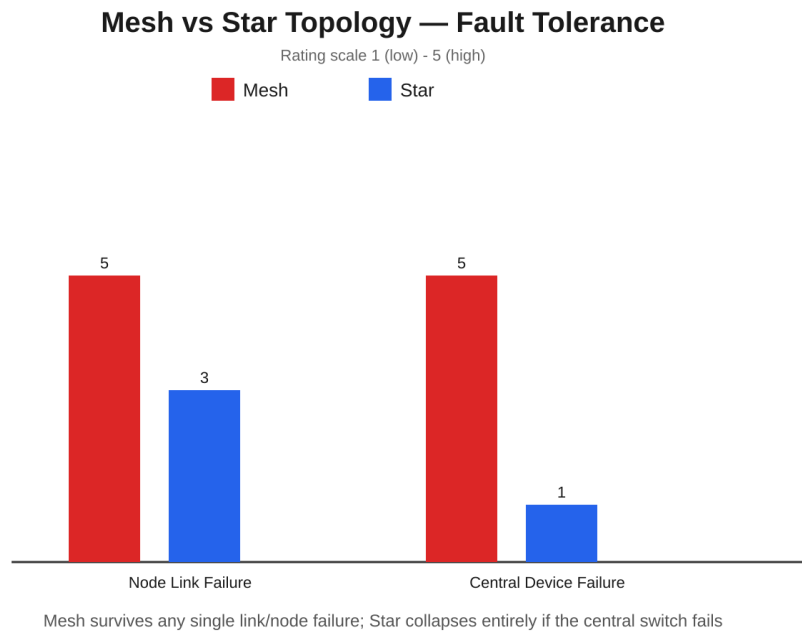


Figure 2: Mesh survives node/link failures; Star fails completely if the central device goes down

3. Bus Topology vs Mesh Topology — Installation Complexity

Understanding of Problem & Constraints

A financial institution handles highly sensitive customer and financial information. Therefore, cybersecurity is a critical requirement.

The organization needs to:

- Protect sensitive customer information.
- Secure online banking services.
- Prevent unauthorized access.
- Comply with strict security policies and regulations.
- Allow legitimate employees to access required resources.
- Maintain high network performance.
- Avoid excessive operational costs.
- Detect suspicious activities.
- Minimize the impact of security incidents.
- Maintain business continuity.

The challenge is to provide strong security without creating excessive latency, complexity, or operational expenses.

Problem Decomposition

The problem can be divided into the following areas:

1. Network segmentation.
2. Firewall deployment.
3. User authentication.
4. Access control.
5. Intrusion detection and prevention.
6. Protection of sensitive databases.
7. Secure online banking infrastructure.
8. Monitoring and logging.
9. Compliance.
10. Performance and cost management.

Application of Constraints in Solution

Network Segmentation

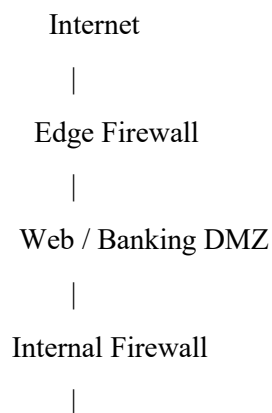
The institution should use **VLANs and security zones** to separate different types of systems.

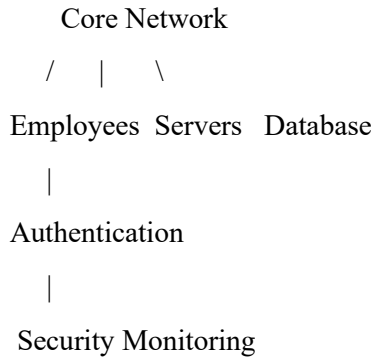
For example:

- Employee network
- Customer-service network
- Server network
- Database network
- Online banking/DMZ network
- Guest network
- Management network

Sensitive databases should not be placed directly on the same network as employee computers.

A simplified structure is:





This segmentation limits the movement of attackers if one system becomes compromised.

Firewall Deployment

The institution should deploy **next-generation firewalls (NGFWs)** at important network boundaries.

The firewall should control:

- Incoming traffic
- Outgoing traffic
- Application traffic
- Unauthorized connections
- Suspicious traffic
- Access between security zones

A DMZ should be used for public-facing services such as online banking web servers.

The database containing sensitive customer information should remain in a protected internal security zone.

Authentication Methods

Strong authentication is essential for financial systems.

The organization should use:

- Multi-factor authentication
- Strong password policies
- Role-based access control
- Centralized identity management
- Least-privilege access
- Privileged access management for administrators

For example, an employee working in customer service should only have access to the customer information required for their job.

An employee should not automatically receive access to databases, administrative systems, or security management systems.

Intrusion Detection Mechanisms

The organization should deploy **IDS/IPS** to detect and potentially block suspicious activity.

An IDS can identify:

- Unusual traffic
- Port scanning
- Malware activity
- Suspicious login behavior
- Known attack patterns

An IPS can actively block or prevent certain detected attacks.

Security logs should be collected centrally so that security administrators can investigate incidents.

A SIEM platform can be used to correlate events from:

- Firewalls
- Servers
- Authentication systems
- IDS/IPS
- Endpoints
- Network devices

Logical Reasoning & Strategy

The primary strategy should be **defense in depth**.

Instead of relying on one security mechanism, multiple layers should protect the institution.

For example:

Internet

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Firewall

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IDS/IPS

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DMZ

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Internal Firewall

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Employee VLAN

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Application Server

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Database VLAN

An attacker would therefore have to overcome multiple security controls to reach sensitive data.

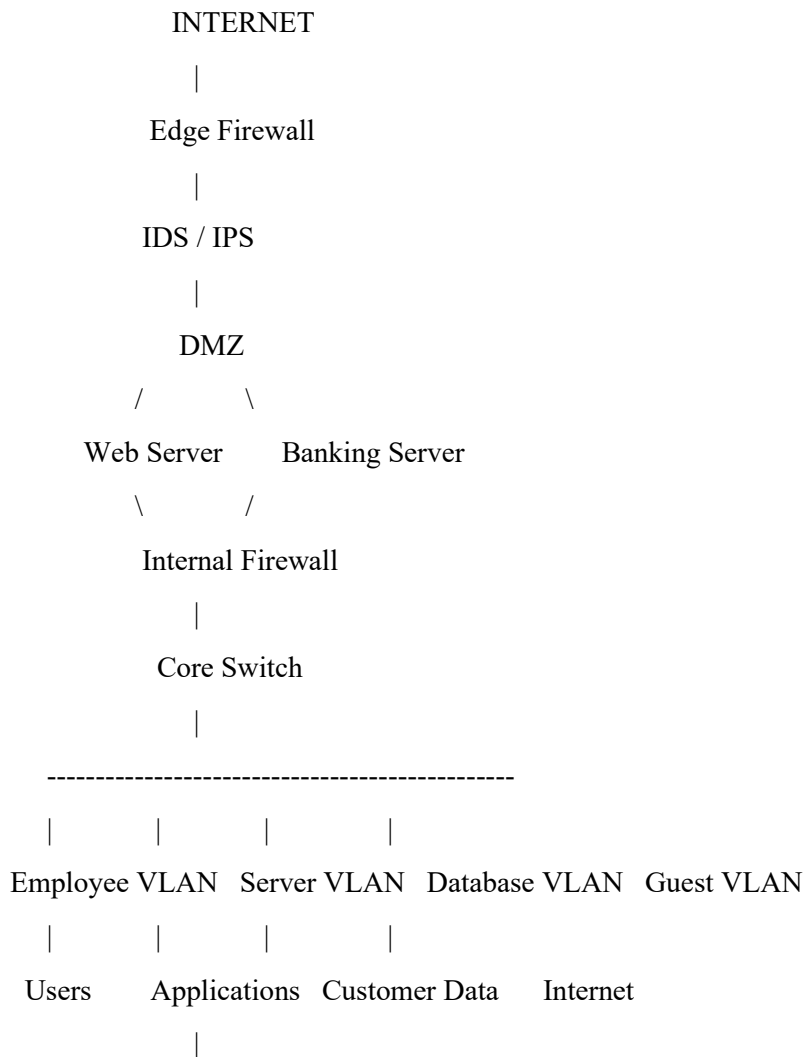
Network segmentation also improves performance because unnecessary broadcast and network traffic can be restricted between different segments.

Role-based access control reduces the number of users who can access sensitive systems.

MFA adds another security layer even if an employee's password is compromised.

Solution Accuracy & Feasibility

A feasible secure architecture is:



SIEM / SOC

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Central Monitoring

The database VLAN should have strict access-control rules. Only authorized application servers should be able to communicate with sensitive database systems.

Guest users should have Internet access but should not be able to access internal financial systems.

Administrative systems should be placed in a separate management network.

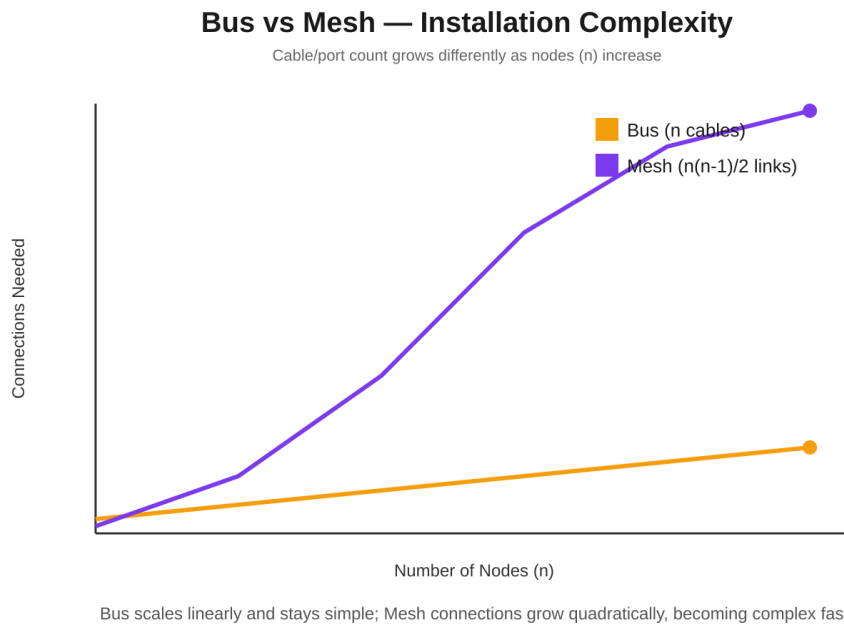


Figure 3: Bus cabling grows linearly with nodes; Mesh connections grow quadratically

4. Hybrid Topology vs Star Topology — Scalability

Understanding of Problem & Constraints

The hospital network has several important constraints that must be considered before designing the solution.

The major constraints are:

- Patient information must remain confidential.
- Network downtime should be minimized.
- Medical applications require reliable connectivity.
- Different departments require different levels of network access.
- The hospital has a limited budget.
- The network should support future expansion.
- Unauthorized users should not access medical records.

- The network should provide sufficient bandwidth for applications.
- Network failures should not completely stop hospital operations.

Therefore, the network must balance **security, reliability, performance, cost, and scalability**.

Problem Decomposition

The problem can be divided into the following sub-problems:

1. Design the network topology.
2. Select suitable networking devices.
3. Separate departments using network segmentation.
4. Protect patient information.
5. Provide reliable Internet connectivity.
6. Provide backup connectivity and power.
7. Control user access.
8. Monitor network activity.
9. Provide wireless access where required.
10. Ensure future scalability.

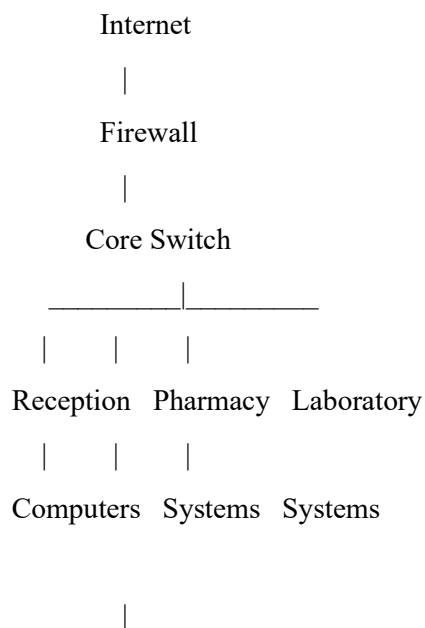
A structured decomposition makes it easier to address each constraint individually.

Application of Constraints in Solution

Network Topology

A **star-based hierarchical topology** is recommended because it is easy to manage and troubleshoot.

Each department can connect to access switches, which connect to core switches.



Medical Servers

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Database Server

This structure allows departments to be managed separately.

Network Segmentation

The hospital should use VLANs to separate departments.

For example:

- VLAN 10 – Reception
- VLAN 20 – Doctors
- VLAN 30 – Laboratory
- VLAN 40 – Pharmacy
- VLAN 50 – Billing
- VLAN 60 – Medical Records
- VLAN 70 – Guest Wi-Fi

This prevents unnecessary communication between departments.

For example, guest Wi-Fi users should not be able to access medical-record servers.

Security

A firewall should be placed between the internal network and the Internet.

Additional security mechanisms should include:

- Multi-factor authentication
- Strong passwords
- Role-based access
- Antivirus/endpoint protection
- IDS/IPS
- Secure Wi-Fi
- Regular backups
- Centralized logging

Medical records should only be accessible to authorized employees.

Logical Reasoning & Strategy

The most important requirement is protecting patient information while keeping medical services operational.

A single flat network would be risky because if one computer becomes compromised, an attacker could potentially access other systems.

Therefore, VLAN segmentation should be used.

For example:

Guest Wi-Fi



Internet Only

Staff Network



Hospital Applications

Medical Records VLAN



Authorized Medical Staff Only

This follows the principle of **least privilege**.

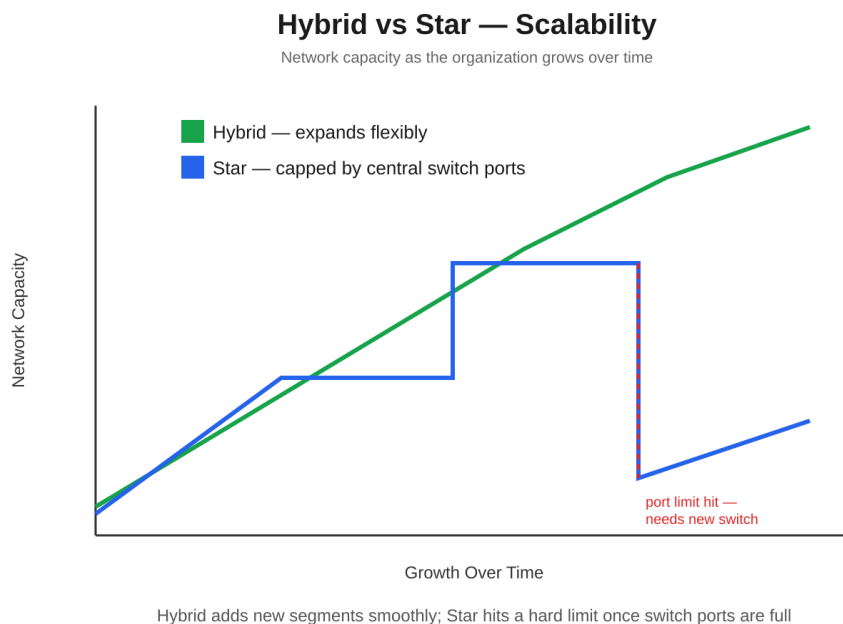


Figure 4: Hybrid expands smoothly; Star growth is capped by central switch port limits

5. Maintenance Comparison — Star, Mesh, Bus, and Hybrid

Understanding of Problem & Constraints

The company has several networking requirements and limitations.

The major constraints are:

Multiple locations must communicate with each other.

Production systems require reliable connectivity.

Cloud applications require Internet access.

Inventory information must be synchronized between locations.

The budget is limited.

Network downtime can affect production.

Sensitive company information must be protected.

Remote branches may not have high-speed wired connections.

The network should support future expansion.

Backup connectivity is required for critical locations.

Therefore, the solution must balance cost, reliability, security, bandwidth, and scalability.

Problem Decomposition

The problem can be divided into:

Connect all company locations.

Select suitable WAN connectivity.

Provide backup connections.

Secure communication between locations.

Design the internal LAN.

Separate production and administrative systems.

Prioritize critical traffic.

Monitor network performance.

Control costs.

Support future branch expansion.

Application of Constraints in Solution

WAN Connectivity

The company can use a combination of:

Fiber broadband where available

4G/5G connectivity for remote branches

Site-to-site VPN

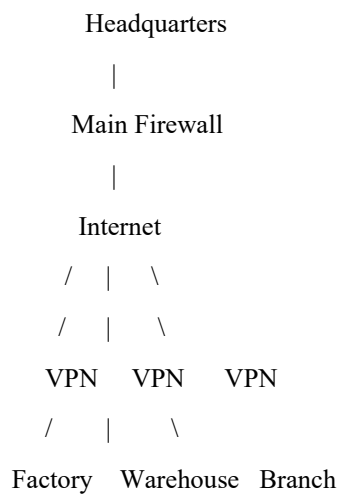
SD-WAN for intelligent traffic management

A practical design would use fiber as the primary connection at locations where it is available and 4G/5G as backup.

Remote branches can use 4G/5G as their primary connection if wired broadband is unavailable.

Site-to-Site VPN

Secure VPN tunnels should connect company locations.



The VPN encrypts traffic between locations and prevents sensitive company data from being transmitted openly.

Network Segmentation

The manufacturing facility should separate different systems.

For example:

Production VLAN

Office VLAN

Management VLAN

Server VLAN

Guest VLAN

IoT/Industrial device VLAN

Production equipment should not be placed on the same unrestricted network as employee computers or guest devices.

This improves both security and network performance.

Maintenance Ease — Star vs Mesh vs Bus vs Hybrid

Rating scale 1 (hard to maintain) - 5 (easy to maintain)

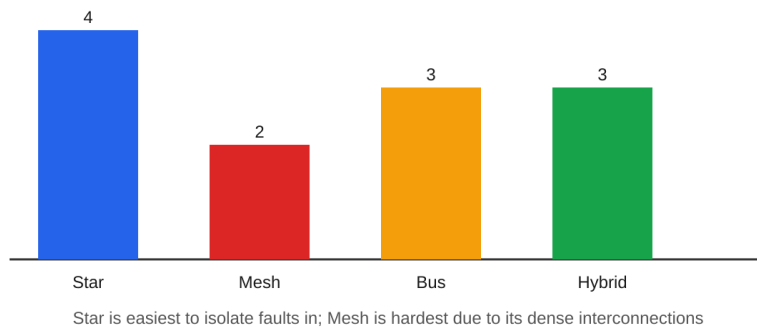


Figure 5: Maintenance ease across all four topologies (higher = easier to maintain)

Rubric for Evaluation

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion